Demand Paging

```
  GNU nano 8.7                                postlab.c                                Modified
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int frames[3];
    int i, j, found;
    int faults = 0, hits = 0;

    // Initialize frames
    for(i = 0; i < 3; i++)
        frames[i] = -1;

    printf("\nDemand Paging Simulation\n");

    for(i = 0; i < n; i++) {
        found = 0;

        // Check if page already in memory
        for(j = 0; j < 3; j++) {
            if(frames[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        // Page Fault
        if(!found) {
            frames[faults % 3] = pages[i]; // Simple replacement
            faults++;
        }

        // Display frames
        printf("Page %d -> ", pages[i]);
        for(j = 0; j < 3; j++)
            printf("%d ", frames[j]);
        printf("\n");

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```
  GNU nano 8.7                                postlab.c

        // Check if page already in memory
        for(j = 0; j < 3; j++) {
            if(frames[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        // Page Fault
        if(!found) {
            frames[faults % 3] = pages[i]; // Simple replacement
            faults++;
        }

        // Display frames
        printf("Page %d -> ", pages[i]);
        for(j = 0; j < 3; j++)
            printf("%d ", frames[j]);
        printf("\n");
    }

    printf("\nTotal Page Faults = %d\n", faults);
    printf("Total Page Hits = %d\n", hits);

    printf("Hit Percentage = %.2f%%\n", (float)hits/n*100);
    printf("Fault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

```
athar@THOR MSYS ~
$ nano postlab.c

athar@THOR MSYS ~
$ gcc postlab.c -o postlab

athar@THOR MSYS ~
$ ./postlab

Demand Paging Simulation
Page 1 -> 1 -1 -1
Page 2 -> 1 2 -1
Page 3 -> 1 2 3
Page 4 -> 4 2 3
Page 1 -> 4 1 3
Page 2 -> 4 1 2
Page 5 -> 5 1 2
Page 1 -> 5 1 2
Page 2 -> 5 1 2
Page 3 -> 5 3 2
Page 4 -> 5 3 4
Page 5 -> 5 3 4

Total Page Faults = 9
Total Page Hits = 3
Hit Percentage = 25.00%
Fault Percentage = 75.00%

athar@THOR MSYS ~
$
```